

```
[3]
✓ 1s
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
```

```
[5]
✓ 0s
data = pd.read_csv("/content/simple_linear_regression.csv")
print(data.head())
```

✓

	ID	Age	Experience	Salary
0	1	36	35	90523
1	2	18	30	65153
2	3	51	12	57843
3	4	55	15	138089
4	5	42	36	88454

```
[6]
✓ 0s
print(data.info())
print(data.describe())
print(data.isnull().sum())
```

✓

```
... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0    ID          500 non-null    int64
1    Age          500 non-null    int64
2    Experience   500 non-null    int64
3    Salary       500 non-null    int64
dtypes: int64(4)
memory usage: 15.8 KB
None
```

	ID	Age	Experience	Salary
count	500.000000	500.000000	500.000000	500.000000
mean	250.500000	38.254000	20.276000	84041.132000
std	144.481833	12.158666	11.488769	37943.453122
min	1.000000	18.000000	0.000000	20017.000000
25%	125.750000	28.000000	11.000000	50287.750000
50%	250.500000	38.000000	20.000000	82764.500000
75%	375.250000	48.000000	30.250000	117583.750000
max	500.000000	60.000000	40.000000	149768.000000
ID	0			
Age	0			
Experience	0			
Salary	0			
dtype:	int64			

```
[17]
✓ 0s
X = data[['Experience']]
y = data['Salary']
```

```
[18]
✓ 0s
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
[19]
✓ 0s
model = LinearRegression()
model.fit(X_train, y_train)
```

✓

LinearRegression

LinearRegression()

[20]
✓ Os

```
y_pred = model.predict(X_test)

print("Predicted Salary:")
print(y_pred[:10])
```

▼

```
Predicted Salary:
[83185.07840822 86141.78035712 83490.94412707 82981.16792899
 84510.49652325 84204.63080439 82573.34697052 84102.67556478
 84612.45176286 82369.43649129]
```

[21]
✓ Os

```
plt.scatter(X_test, y_test, color='blue')
plt.plot(X_test, y_pred, color='red')
plt.xlabel("Experience")
plt.ylabel("Salary")
plt.title("Simple Linear Regression")
plt.show()
```

...

